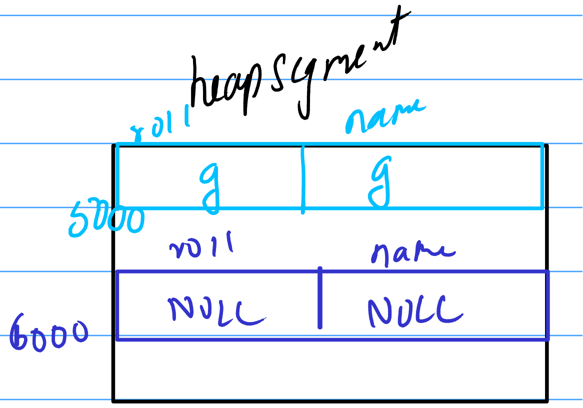
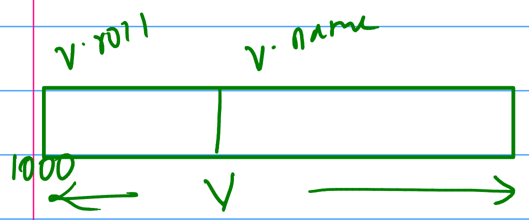
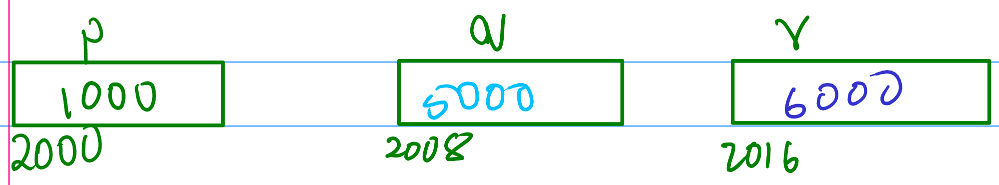
26/03

```
struct  Student
{               int  roll;
                char  name[20];
};
struct  Student  V;
struct  Student  *p,*q,*r;
p=&v;
q=malloc(sizeof(struct  Student));
r=calloc(1,sizeof(struct  Student));
```







v.roll
v.name

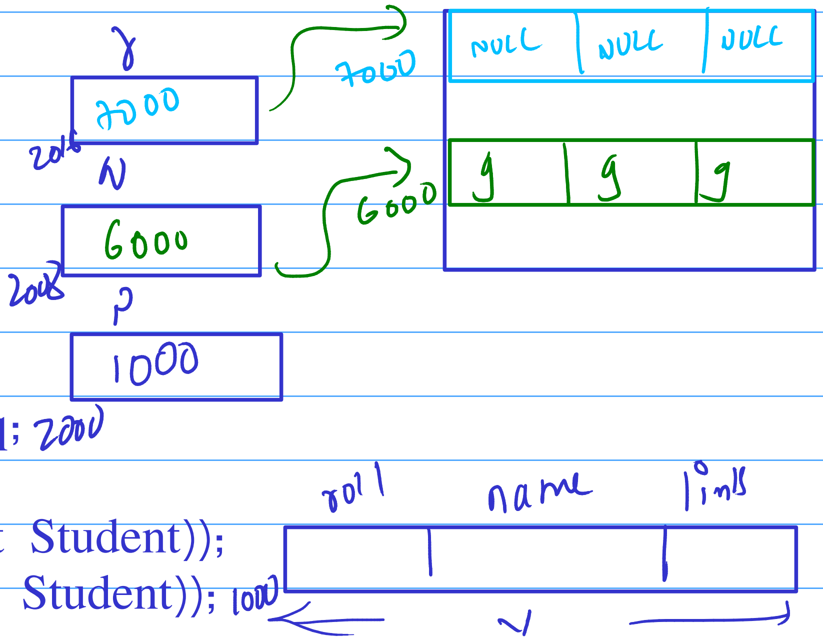
p->roll
p->name

q->roll
q->name

r->roll
r->name

```
struct  Student
{  int  roll;
char  name[20];
struct  Student  *link;
};

struct  Student  v;
struct  Student  *p,*r,*q;
p=&v;
r=malloc(sizeof(struct  Student));
q=calloc(sizeof(struct  Student));
```



v.roll, v.name, v.lin^{6}s

p→roll, p→name, p→lin^{6}s

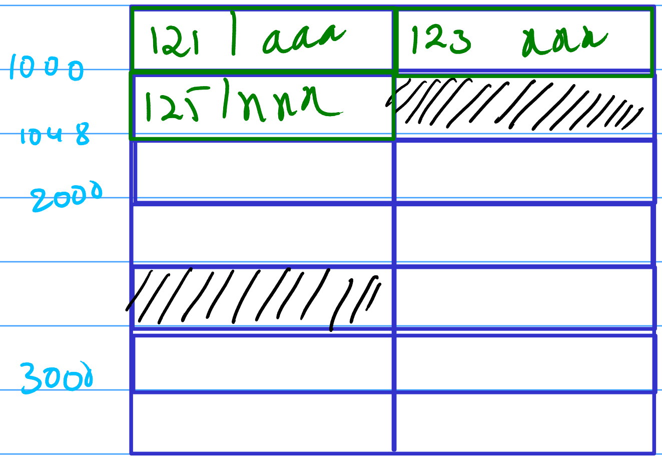
q→roll, ~~a~~→name, q→lin^{6}s

same with ~~a~~ .

DB
$\boxed{\text{~~roll~~ 1000}}$

ct
$\boxed{\cancel{0}\ \cancel{1}\ \cancel{2}\ 3}$

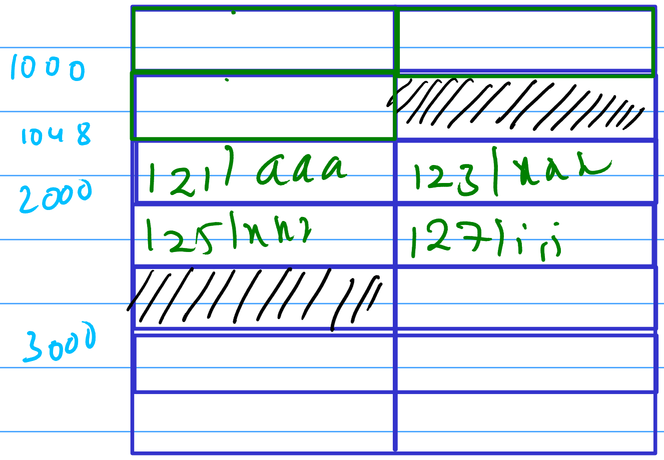
DB = ADD(DB);
DB = ADD(DB);
DB = ADD(DB);
DB = ADD(DB);
↳ next contiguous memory is not free so realloc will search x(ct+1) mem.



DB
$\boxed{2000}$

ct
$\boxed{\cancel{3}\ 4}$

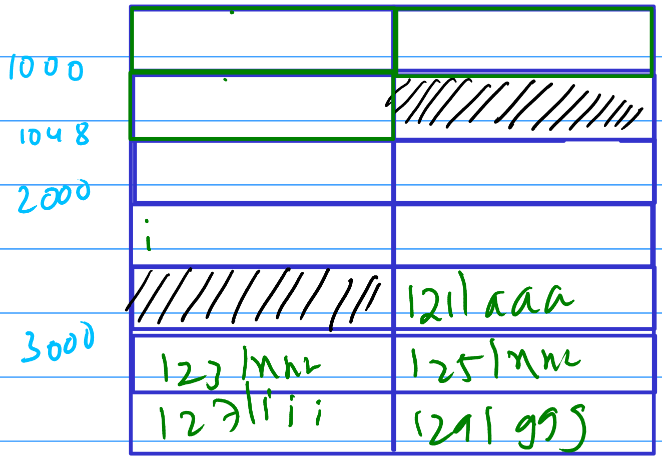
DB = ADD(DB);



DB
$\boxed{3000}$

ct
$\boxed{4}$

DB = ADD(DB)
↳ failed to alloc mem



value of x

```
int  x=10;
int  a[]={10,20,30,40};
```

element of a

```
struct  St
{               int  x;
                int  y;
}v={10,20};
```

member of Struct

```
struct  St
{
                int  id;
                char  name[20];
};
struct  St  A[5];
```

A[0] → record.

```
struct  St
{
                int  roll;
                char  name[20];
                struct  St  *link;

};
```

node

Datastructures

What is Datastructures ?

Organisation of data in the memory in some systematic order , that accessing beoming faster

Types of datastructures :

linear datastructures

non linear datastructures

linear datastructures : in linear datastructures data is accessed in a sequential order.
examples of linear datastructures :
1. Array
2. link list
3. stack
4. queue

Note : data accessed ina sequentail order , data need not store i
a contuguous locations

Nonlinear datastructures : in nonlinear datastructures data is accessed in a nonsequential order.
Examples of Nonlinear datastructures:
1.trees → Binary Search Tree
2. graphs

TimeComplexity : time taken to execute the task is called time complexity
time complexity is divided into 3 categories
best case : minimum time taken
average case : medium time taken
worst case : maximum time taken

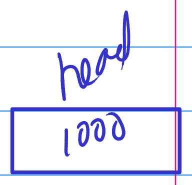
SpaceComplexity : memory required to run the task is called spacecomplexity

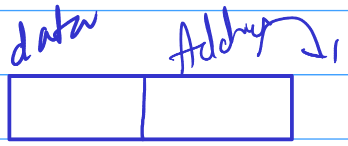
timecomplexity and spacecomplexity are represented with
Big O notaion ---> O()

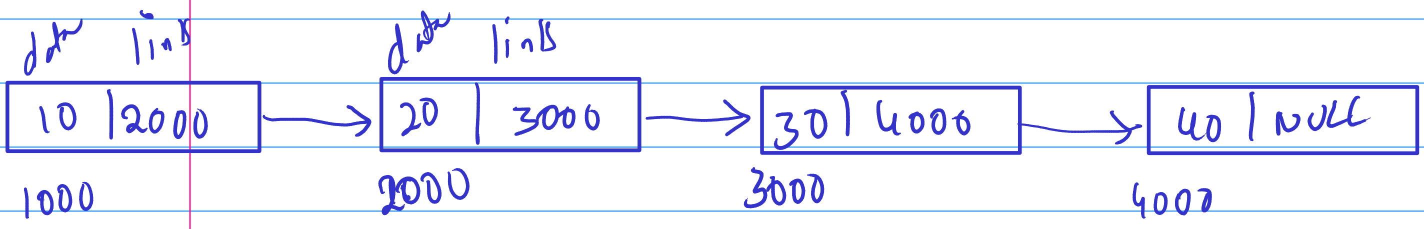
frquently used Big notations

O(1) ----> constant time or constant memory
O(n) ----> directly proportional to n
O(2n) ----> direclty proportional to twice the n value
O(n_2) ---> directly proportional to square of n
O(logn)--> directly logarithm of n

Link List : Link list is a collection of nodes, where every no
is a combination of current node information and next node
address







types of link list
1. singly link list
2. double link list
3. circular link list